Qiyao Xue

PhD Student in Data Science, University of North Carolina at Chapel Hill

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Research Interests

Current: Agentic and self-evolving AI: agents that reason, plan, use tools, and improve from feedback.
Previous: Multimodal generative AI, visual spatial reasoning, and physics-grounded video generation.

Education

University of North Carolina at Chapel Hill — Chapel Hill, NC
Ph.D. in Data Science — *Aug 2026 – Present*

University of Pittsburgh — Pittsburgh, PA
Ph.D. student in Electrical and Computer Engineering (transferred to UNC) — *Aug 2024 – Jul 2026*

The Hong Kong Polytechnic University — Hong Kong
B.Eng. in Electrical Engineering, First Class Honours; Minor in Applied Mathematics — *Sep 2019 – Jun 2023*

- GPA 3.74/4.3 (top 2%); exchange at University of Maryland, College Park (Spring 2022).

Publications

Conference & Workshop Papers

- **PhyT2V: LLM-Guided Iterative Self-Refinement for Physics-Grounded Text-to-Video Generation**
  CVPR, 2025
  Qiyao Xue, Xiangyu Yin, Boyuan Yang, Wei Gao
- **MMBERT: Scaled Mixture-of-Experts Multimodal BERT for Robust Chinese Hate Speech Detection under Cloaking Perturbations**
  AAAI, 2026
  Qiyao Xue, Yuchen Dou, Ryan Shi, Xiang Lorraine Li, Wei Gao
- **Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective**
  ECCV, 2026
  Qiyao Xue, Weichen Liu, Shiqi Wang, Haoming Wang, Yuyang Wu, Wei Gao
- **ProGait: A Multi-Purpose Video Dataset and Benchmark for Transfemoral Prosthesis Users**
  ICCV, 2025. **Highlight**
  Xiangyu Yin, Boyuan Yang, Weichen Liu, **Qiyao Xue**, Abrar Alamri, Goeran Fiedler, Wei Gao
- **InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity**
  CVPR, 2026. **Oral**
  Haoming Wang, **Qiyao Xue**, Wei Gao
- **SpatialMind: Spatially Aware On-Device Embodied AI via Viewpoint Integration**
  ACM MobiCom, 2026
  Haoming Wang, **Qiyao Xue**, Weichen Liu, Boyuan Yang, Xiangyu Yin, Wei Gao
- **KGRxn-LLM: Knowledge Graph Enhanced Large Language Models for Molecular Reaction Reasoning**
  BioNLP Workshop @ ACL, 2026
  Weichen Liu, **Qiyao Xue**, Yuyang Wu, Olexandr Isayev, Natasa Miskov-Zivanov

Preprints

- **CLORE: Content-Level Optimization for Reasoning Efficiency**
  arXiv:2605.22211, 2026
  Yuyang Wu, **Qiyao Xue**, Guanxing Lu, Weichen Liu, Zihan Wang, Manling Li, Olexandr Isayev
- **Can Agents Price a Reaction? Evaluating LLMs on Chemical Cost Reasoning**
  arXiv:2605.07251, 2026
  Yuyang Wu, Yue Huang, Shuaike Shen, Xujian Wang, Shuhao Zhang, **Qiyao Xue**, Weichen Liu, Runtian Gao, Jian Ma, Xiangliang Zhang, Olexandr Isayev
- **Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods**
  arXiv:2511.15722, 2025
  Weichen Liu, **Qiyao Xue**, Haoming Wang, Xiangyu Yin, Boyuan Yang, Wei Gao

Research Experience

Intelligent Systems Lab, University of Pittsburgh — Pittsburgh, PA
Graduate Research Assistant, with Prof. Wei Gao — *Aug 2024 – Jul 2026*

- Proposed PhyT2V, a training-free, LLM-guided self-refinement loop that improves the physical realism and temporal causality of text-to-video diffusion models (CVPR 2025, first author).
- Built InfiniBench, a procedurally generated benchmark for visual spatial reasoning with controllable scene complexity (CVPR 2026 Oral), and analyzed MLLM reasoning paths and latent states on multi-view spatial tasks (ECCV 2026, first author).
- Designed MMBERT, a Mixture-of-Experts multimodal BERT with a three-stage training recipe for robust hate speech detection under adversarial cloaking (AAAI 2026, first author).

Med-AIR Lab, The Chinese University of Hong Kong — Hong Kong
Research Assistant, with Prof. Qi Dou — *Jul 2023 – Mar 2024*

- Developed nnU-Net / Transformer-based models for multi-organ segmentation and ViT-based multi-disease detection on medical images, and a SAM-based interactive 3D MRI annotation tool.

Honors & Awards

Outstanding Student Award (top 1 in department), Dept. of Electrical Engineering, PolyU	2023
President Emeritus Professor Poon Chung-kwong Scholarship (top 2), PolyU	2022
Wong Ti-shing Student Exchange Scholarship, PolyU	2022
Undergraduate Research Innovation Scheme Scholarship, PolyU	2022
HKSAR Government Scholarship Fund – Reaching Out Award	2022
VTech Group of Companies Scholarship (top 4), PolyU	2022
Professor Leung Tin-pui Memorial Scholarship (top 2), PolyU	2021
Dean's Honours List, PolyU	2020 – 2022

Skills

Programming: Python (PyTorch, Hugging Face Transformers/Diffusers, NumPy, pandas), C/C++, MATLAB, SQL
Tools: Linux, Git, Docker, LaTeX